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Survey on the Needs for Chemistry Research Data Management and Sharing

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ABSTRACT

This paper aims to reveal the situation of research data in chemistry research process and chemistry researchers' need for data management support from five perspectives, i.e., data generation and collection, data recording and processing, data preservation and backup, data publication and sharing, needs for data management and sharing services. Our survey is based on a questionnaire carried out among 119 subjects, i.e., researchers and graduate students in chemistry of Chinese Academy of Science. The analysis results provide us with a better understanding on the current attitudes and needs of researchers and graduate students about data management and sharing in chemistry. Although this survey was implemented in chemistry, it could provide us with some inspirations for designing a range of library services for other disciplines, particularly in promotion, consulting and training of research data management and sharing, and research data storage.

Introduction

In the context of e-science and open data, some governments, international organizations, funding agencies and periodical publishers have issued a series of research data management and sharing (RDMS) policies in recent years. Researchers are encouraged or required to share research data along with their research outputs. These open data policies have actively promoted research data sharing and reusing during research data lifecycle.

In reality, researchers not only have different attitudes and expectancies for managing and sharing research data (Carlson & Stowell-Bracke, 2013; Hall, 2013; Huang et al., 2012; Tenopir et al., 2011), but also meet many challenges in management data especially for different disciplines of small science (Borgman, 2012; Borgman, Wallis, & Enyedy, 2007; Gray et al., 2005). During the past few years, issues relating to research data management in general and libraries' data management services for researchers in particular, have attracted a great deal of attention in the library community, so that a number of guides or handbooks (e.g., The Medical Library Association Guide to Data Management for Librarians, Data Management: A Practical Guide for Librarians, Research Data Management: Practical Strategies for Information Professionals, etc.) have been published. However, these practices are not adequate to inform libraries in their design of data management services for researchers. To make sure that libraries can provide personalized, specific and effective services for researchers, it is necessary to understand the current situation of research data management and needs for RDMS services in the research process. Therefore some studies were conducted to survey on the current states and needs of researchers and graduate students for RDMS(Anderson et al., 2007; Carlson, 2012; Peters & Dryden, 2011), and others are increasingly engaged in understanding practices and developing new services to satisfy research data management requirements, particularly within academic library(Bardyn, Resnick, & Camina, 2012; Kim, 2015; Williams, 2013).

National Science Library (NSL) is an academic library attached to Chinese Academy of Sciences (CAS). In recent years, NSL has initiated subject librarian system and set up full time subject librarian's team with diverse subject backgrounds. Providing subject information services for the researchers and graduates of the institutes of CAS in different research fields are one important aspect of subject librarians' job. With RDMS becoming more and more important, many researchers also have shown much concern about the issue. So in this study, we conducted the survey for chemistry researchers and graduates in Chemistry Institute of CAS from five perspectives of research data in chemistry research process, i.e., data generation and collection, data recording and processing, data preservation and backup, data publication and sharing, needs for data management and sharing services. By means of questionnaires survey, we collected solid evidence and raw data on RDMS in chemistry. Based on the statistical analytical results, we have a better understanding on their current attitudes and needs of researchers about their data management and sharing. Although the survey was implemented in chemistry, it could be a reference for us to design a

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